

HCSM-27: Klein Orbit Structure

The Klein Four-Group Action, Kernel Classification, and the Framework-Constant Four-Cycle Orbit

Stanley Preschutti^{*1}

¹Entropia Research Institute / Information Physics Institute

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Abstract

The Hodge Complex Standard Model (HCSM) derives the Standard Model of particle physics from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation.

Two order-two operations act naturally on the fourteen-mode multiplet: the diagonal reflection rs , identified in HCSM-26 as the unique chiral reflection of the D_4 point group, and the half-period translation T_{half} . They commute, and each squares to the identity, so they generate a Klein four-group $V = \langle rs, T_{\text{half}} \rangle$.

This paper proves four results. The *Klein Orbit Theorem*, the *Kernel Classification Theorem*, and the *Swap-Pair Bridge Theorem* are fully derived from the substrate at coordinate level. The *Four-Orbit Framework-Constant Theorem* is derived from the intra-pair winding-matrix magnitudes, which are co-ordinate arithmetic on the four-cycle orbit.

The Klein four-group acts on the fourteen-mode multiplet with five size-2 orbits and one size-4 orbit. The pointwise stabilizers of these orbits are the three non-trivial subgroups of the Klein group. One orbit — the four-cycle orbit $V_6 = \{(5, 7), (7, 5), (1, 3), (3, 1)\}$ — is the framework-constant orbit. Its four $N/4$ anchors are the framework constants $(d-1)^2$, $(d-1)d$, $2(L-1)$, and $(d-1)(2d-1)$. They satisfy two additive identities, sum to $56 = \dim V_{56}$, and have mean $14 = 2(L-1)$.

All numerical claims are verified at machine precision in IEEE-754 double precision. The complete verification script is provided.

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^{*}ORCID: 0009-0004-5445-1744. Email: scstannp@yahoo.com

1 Introduction

1.1 The Klein group in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2]. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation. Their D_4 representation theory determines the gauge algebra [4, 5]; their $N/4$ labels determine the mass spectrum [13, 16]; the zero-state vector from the 14×19 measurement matrix determines the flavor structure [12, 14].

Two order-two operations act on the fourteen-mode multiplet. The first is the diagonal reflection rs , which swaps the two mode coordinates and has been identified in HCSM-26 [15] as the unique chiral reflection of the D_4 point group. The second is the half-period translation T_{half} , which shifts both mode coordinates by half the torus side length. The two operations commute, and each squares to the identity, so they generate a Klein four-group.

This paper proves that the action of the Klein group organizes the fourteen-mode multiplet completely: the orbits are classified, the stabilizers of the orbits are classified, and one orbit in particular — the four-cycle orbit — contains four $N/4$ anchors whose values are all framework constants.

1.2 One-line summary

The Klein four-group $V = \langle rs, T_{\text{half}} \rangle$ acts on the fourteen-mode multiplet with five size-2 orbits and one size-4 orbit. The pointwise stabilizers of these orbits fall into the three non-trivial subgroups of the Klein group. The four-cycle orbit is the framework-constant orbit: its four $N/4$ anchors are simultaneously framework constants and sum to 56 = $\dim V_{56}$.

1.3 Derivation status

The paper proves four results. They split into two groups.

Fully derived (no anchors).

- (1) **Klein Orbit Theorem** (Theorem 3.2): the Klein four-group acts on the fourteen-mode multiplet with orbit decomposition consisting of five size-2 orbits and one size-4 orbit.
- (2) **Kernel Classification Theorem** (Theorem 4.2): the pointwise stabilizer of each Klein orbit falls into one of the three non-trivial subgroups of the Klein group.
- (3) **Swap-Pair Bridge Theorem** (Theorem 5.2): the edge $\{(5, 7), (6, 6)\}$ is the unique edge in the distinguished-pair graph that connects two distinct Klein orbits.

Derived from the intra-pair winding-matrix magnitudes.

- (4) **Four-Orbit Framework-Constant Theorem** (Theorem 6.6): the four-cycle orbit's four $N/4$ anchors are simultaneously framework constants. They satisfy two additive identities, sum to 56, and have mean 14.

1.4 Organization

Section 2 states the setup. Section 3 proves the Klein Orbit Theorem. Section 4 proves the Kernel Classification Theorem. Section 5 proves the Swap-Pair Bridge Theorem. Section 6 proves the Four-Orbit Framework-Constant Theorem. Section 7 discusses the structural interpretation. Section 8 catalogs inputs and derived results. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup

2.1 The substrate

Axiom 2.1 (Lattice primitive). There exists a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L a positive even integer.

The substrate is the periodic eight-by-eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = L^2 = 64$ sites. The framework constants are

$$d = 4, \quad N = 64, \quad L = 8, \quad H = 2^d = 16. \quad (1)$$

The discrete five-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x_1 + e_1, x_2 + e_2) \quad (2)$$

has spectrum

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right) \quad (3)$$

for $(n_1, n_2) \in \{0, \dots, 7\}^2$.

2.2 The fourteen-mode multiplet

The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L-1) = 14$. The fourteen modes decompose under the D_4 point group of the square into three orbits:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (4)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (5)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (6)$$

The orbit sizes are $(2, 8, 4)$.

In the standard mode order used throughout HCSM,

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (7)$$

the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (8)$$

2.3 The D_4 action

The D_4 point group is generated by the rotation

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8} \quad (9)$$

and the reflection

$$s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8}, \quad (10)$$

subject to the presentation $r^4 = s^2 = (rs)^2 = e$. The five order-two elements are r^2 , s , rs , r^2s , r^3s . The diagonal reflection is

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (11)$$

2.4 The winding generator

The winding generator is

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8}M\right), \quad M_{ij} = n_1^{(i)}n_2^{(j)} - n_1^{(j)}n_2^{(i)}. \quad (12)$$

Its spectrum is

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (13)$$

2.5 The processor

The processor is the rank-10 orthogonal projection

$$T = I - P_C, \quad C = \{\text{dark}, W, Z, H\}. \quad (14)$$

Its image is $F = A \cup B_{\text{class}}$ with $|F| = 10$; its kernel is $B_{\text{box}} = C$ with $|B| = 4$.

2.6 The zero state

A 14×19 measurement matrix of vacuum-native observables has rank 13 and nullity 1 [12]. Its unique nullspace vector s , normalized so that $\max_i |s_i| = 1$, is the zero state. In the standard mode order:

$$\begin{aligned} s = & (-0.105198199128, +0.105198199128, +0.937726561300, \\ & -0.192806690737, -0.937726561300, +0.658283218429, \\ & -1.000000000000, -0.658283218429, +1.000000000000, \\ & +0.192806690737, 0.000000000000, -0.329141609215, \\ & +0.329141609215, 0.000000000000). \end{aligned} \quad (15)$$

The two coordinates with $s = 0$ exactly are (2, 2) and (6, 6).

2.7 The half-period translation

Definition 2.2 (Half-period translation). The half-period translation is the map

$$T_{\text{half}} : (n_1, n_2) \mapsto (n_1 + 4 \pmod{8}, n_2 + 4 \pmod{8}). \quad (16)$$

Lemma 2.3 (T_{half} is an involution). $T_{\text{half}}^2 = \text{id}$.

Proof. $(n_1 + 4 + 4) \pmod{8} = n_1 \pmod{8}$, and likewise for n_2 . $\square$

2.8 Inputs

The paper uses the following inputs from the HCSM stack.

- **Input 1.** The substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by the unified substrate selection theorem [1].
- **Input 2.** The fourteen-mode multiplet at $\lambda = 4$ [3].
- **Input 3.** The D_4 action on the fourteen modes [4].
- **Input 4.** The winding generator H_{wind} and its spectrum [9].
- **Input 5.** The processor T [8].
- **Input 6.** The zero state s [12].

3 The Klein Orbit Theorem

3.1 The Klein four-group

Lemma 3.1 (Klein structure). *The diagonal reflection rs and the half-period translation T_{half} commute, and each squares to the identity:*

$$rs^2 = e, \quad T_{\text{half}}^2 = e, \quad [rs, T_{\text{half}}] = 0. \quad (17)$$

Hence they generate a Klein four-group

$$V = \langle rs, T_{\text{half}} \rangle = \{e, rs, T_{\text{half}}, rs \cdot T_{\text{half}}\}. \quad (18)$$

Proof. The relations $rs^2 = e$ and $T_{\text{half}}^2 = e$ are immediate from Equation (11) and Equation (16). For the commutator, let (n_1, n_2) be any mode. Then

$$\begin{aligned} rs(T_{\text{half}}(n_1, n_2)) &= rs(n_1 + 4, n_2 + 4) = (n_2 + 4, n_1 + 4), \\ T_{\text{half}}(rs(n_1, n_2)) &= T_{\text{half}}(n_2, n_1) = (n_2 + 4, n_1 + 4). \end{aligned}$$

The two expressions coincide, so $[rs, T_{\text{half}}] = 0$. The group generated by two commuting involutions is the Klein four-group. $\square$

3.2 Orbit decomposition

Theorem 3.2 (Klein Orbit Theorem). *The Klein four-group $V = \langle rs, T_{\text{half}} \rangle$ acts on the fourteen-mode multiplet with orbit decomposition consisting of five orbits of size 2 and one orbit of size 4. Explicitly, the orbits are*

$$V_1 = \{(0, 4), (4, 0)\}, \quad (19)$$

$$V_2 = \{(6, 2), (2, 6)\}, \quad (20)$$

$$V_3 = \{(1, 5), (5, 1)\}, \quad (21)$$

$$V_4 = \{(3, 7), (7, 3)\}, \quad (22)$$

$$V_5 = \{(2, 2), (6, 6)\}, \quad (23)$$

$$V_6 = \{(5, 7), (7, 5), (1, 3), (3, 1)\}. \quad (24)$$

The total mode count is $5 \cdot 2 + 1 \cdot 4 = 14$, matching the multiplicity at the self-paired eigenvalue.

Proof. Direct computation of the orbit of each coordinate under the Klein group. We compute representative orbits.

Orbit of $(0, 4)$:

$$\begin{aligned} e \cdot (0, 4) &= (0, 4), & rs \cdot (0, 4) &= (4, 0), \\ T_{\text{half}} \cdot (0, 4) &= (4, 0), & (rsT_{\text{half}}) \cdot (0, 4) &= (0, 4). \end{aligned}$$

The orbit is $\{(0, 4), (4, 0)\}$, size 2.

Orbit of $(6, 2)$:

$$\begin{aligned} e \cdot (6, 2) &= (6, 2), & rs \cdot (6, 2) &= (2, 6), \\ T_{\text{half}} \cdot (6, 2) &= (2, 6), & (rsT_{\text{half}}) \cdot (6, 2) &= (6, 2). \end{aligned}$$

The orbit is $\{(6, 2), (2, 6)\}$, size 2.

Orbit of $(1, 5)$:

$$\begin{aligned} e \cdot (1, 5) &= (1, 5), & rs \cdot (1, 5) &= (5, 1), \\ T_{\text{half}} \cdot (1, 5) &= (5, 1), & (rsT_{\text{half}}) \cdot (1, 5) &= (1, 5). \end{aligned}$$

The orbit is $\{(1, 5), (5, 1)\}$, size 2.

Orbit of $(3, 7)$:

$$\begin{aligned} e \cdot (3, 7) &= (3, 7), & rs \cdot (3, 7) &= (7, 3), \\ T_{\text{half}} \cdot (3, 7) &= (7, 3), & (rsT_{\text{half}}) \cdot (3, 7) &= (3, 7). \end{aligned}$$

The orbit is $\{(3, 7), (7, 3)\}$, size 2.

Orbit of $(2, 2)$:

$$\begin{aligned} e \cdot (2, 2) &= (2, 2), & rs \cdot (2, 2) &= (2, 2), \\ T_{\text{half}} \cdot (2, 2) &= (6, 6), & (rsT_{\text{half}}) \cdot (2, 2) &= (6, 6). \end{aligned}$$

The orbit is $\{(2, 2), (6, 6)\}$, size 2.

Orbit of $(5, 7)$:

$$\begin{aligned} e \cdot (5, 7) &= (5, 7), & rs \cdot (5, 7) &= (7, 5), \\ T_{\text{half}} \cdot (5, 7) &= (1, 3), & (rsT_{\text{half}}) \cdot (5, 7) &= (3, 1). \end{aligned}$$

The orbit is $\{(5, 7), (7, 5), (1, 3), (3, 1)\}$, size 4.

The six orbits cover all fourteen coordinates. $\square$

Table 1: Pointwise stabilizers of the Klein orbits.

Orbit	Members	Pointwise stabilizer
V_1	$\{(0, 4), (4, 0)\}$	$\{e, rsT_{\text{half}}\}$
V_2	$\{(6, 2), (2, 6)\}$	$\{e, rsT_{\text{half}}\}$
V_3	$\{(1, 5), (5, 1)\}$	$\{e, rsT_{\text{half}}\}$
V_4	$\{(3, 7), (7, 3)\}$	$\{e, rsT_{\text{half}}\}$
V_5	$\{(2, 2), (6, 6)\}$	$\{e, rs\}$
V_6	$\{(5, 7), (7, 5), (1, 3), (3, 1)\}$	$\{e\}$

3.3 The three structural regions

The orbit decomposition partitions the fourteen-mode multiplet into three structurally distinct regions:

1. The four two-cycle orbits V_1 – V_4 . Each orbit has two members exchanged by rs and by T_{half} , and fixed by rsT_{half} .
2. The observer orbit $V_5 = \{(2, 2), (6, 6)\}$. Both members are fixed by rs .
3. The four-cycle orbit $V_6 = \{(5, 7), (7, 5), (1, 3), (3, 1)\}$. The Klein group acts as a 4-cycle.

Remark 3.3. The four-cycle orbit V_6 contains the coordinates $(5, 7), (7, 5), (1, 3), (3, 1)$, all drawn from the middle D_4 orbit $\mathcal{O}_1$. The Klein group therefore refines $\mathcal{O}_1$: the D_4 orbit of size 8 splits into the three Klein orbits $V_3 = \{(1, 5), (5, 1)\}$, $V_4 = \{(3, 7), (7, 3)\}$, and $V_6 = \{(5, 7), (7, 5), (1, 3), (3, 1)\}$.

4 Kernel Classification

Definition 4.1 (Pointwise stabilizer). For a Klein orbit $V_i \subset \mathcal{M}$, the pointwise stabilizer is the subgroup of V consisting of elements g such that $g \cdot m = m$ for every mode $m \in V_i$.

Theorem 4.2 (Kernel Classification Theorem). *The pointwise stabilizer of each Klein orbit is one of the three non-trivial subgroups of the Klein group. The classification is given in Table 1.*

Proof. Direct computation of the action of each group element on each orbit. We compute representative cases.

Orbit $V_1 = \{(0, 4), (4, 0)\}$:

$$\begin{aligned} rs \cdot (0, 4) &= (4, 0) \neq (0, 4) && \text{(swaps)} \\ T_{\text{half}} \cdot (0, 4) &= (4, 0) \neq (0, 4) && \text{(swaps)} \\ (rsT_{\text{half}}) \cdot (0, 4) &= (0, 4), & (rsT_{\text{half}}) \cdot (4, 0) &= (4, 0) \quad \text{(fixed pointwise)} \end{aligned}$$

The pointwise stabilizer is $\{e, rsT_{\text{half}}\}$.

Orbit $V_5 = \{(2, 2), (6, 6)\}$:

$$\begin{aligned} rs \cdot (2, 2) &= (2, 2), & rs \cdot (6, 6) &= (6, 6), \\ T_{\text{half}} \cdot (2, 2) &= (6, 6) \neq (2, 2), \\ (rsT_{\text{half}}) \cdot (2, 2) &= (6, 6) \neq (2, 2). \end{aligned}$$

The pointwise stabilizer is $\{e, rs\}$.

Orbit $V_6 = \{(5, 7), (3, 1), (1, 3), (7, 5)\}$:

$$\begin{aligned} rs \cdot (5, 7) &= (7, 5) \neq (5, 7), \\ T_{\text{half}} \cdot (5, 7) &= (1, 3) \neq (5, 7), \\ (rsT_{\text{half}}) \cdot (5, 7) &= (3, 1) \neq (5, 7). \end{aligned}$$

No non-identity element fixes every member of V_6 . The pointwise stabilizer is $\{e\}$.

The remaining orbits are computed analogously. $\square$

Corollary 4.3 (Three stabilizer classes). *The Klein orbits partition into three stabilizer classes:*

- Class $\{e, rsT_{\text{half}}\}$: four orbits, V_1, V_2, V_3, V_4 .
- Class $\{e, rs\}$: one orbit, V_5 .
- Class $\{e\}$: one orbit, V_6 .

Proof. Immediate from Theorem 4.2. $\square$

Remark 4.4. The observer orbit V_5 is the unique orbit whose pointwise stabilizer is generated by the chiral reflection rs . This is consistent with the Chirality Selection Theorem of HCSM-26 [15]: the coordinate pair $\{(2, 2), (6, 6)\}$ is the unique fixed locus of rs on the fourteen-mode multiplet.

5 The Swap-Pair Bridge Theorem

Definition 5.1 (Distinguished-pair graph). The distinguished-pair graph of the fourteen-mode multiplet has a vertex for each mode and an edge for each two-element subset that is distinguished by at least one of the following structural criteria:

- (i) being fixed by an involution of the D_4 point group;
- (ii) being a two-cycle of the diagonal reflection rs ;
- (iii) being a two-cycle of the half-period translation T_{half} ;
- (iv) being the orbit-class swap pair.

Theorem 5.2 (Swap-Pair Bridge Theorem). *At coordinate level, the edge $\{(5, 7), (6, 6)\}$ is the unique edge in the distinguished-pair graph of the fourteen-mode multiplet that connects two distinct Klein orbits.*

Proof. The ten edges of the distinguished-pair graph are listed in Table 2. Nine edges have both endpoints in the same Klein orbit:

$$\begin{aligned} \{(1, 3), (3, 1)\}, \{(1, 3), (5, 7)\}, \{(3, 1), (7, 5)\}, \{(5, 7), (7, 5)\} &\subset V_6, \\ \{(0, 4), (4, 0)\} &\subset V_1, \\ \{(1, 5), (5, 1)\} &\subset V_3, \\ \{(3, 7), (7, 3)\} &\subset V_4, \\ \{(2, 6), (6, 2)\} &\subset V_2, \\ \{(2, 2), (6, 6)\} &\subset V_5. \end{aligned}$$

Table 2: Distinguished-pair graph edges at coordinate level.

Index	Edge (coordinates)
1	$\{(0, 4), (4, 0)\}$
2	$\{(1, 3), (3, 1)\}$
3	$\{(1, 3), (5, 7)\}$
4	$\{(1, 5), (5, 1)\}$
5	$\{(3, 7), (7, 3)\}$
6	$\{(3, 1), (7, 5)\}$
7	$\{(5, 7), (7, 5)\}$
8	$\{(5, 7), (6, 6)\}$
9	$\{(2, 6), (6, 2)\}$
10	$\{(2, 2), (6, 6)\}$

The remaining edge is $\{(5, 7), (6, 6)\}$. Its two endpoints are in distinct Klein orbits:

$$(5, 7) \in V_6, \quad (6, 6) \in V_5.$$

Hence $\{(5, 7), (6, 6)\}$ is the unique inter-orbit edge. $\square$

Corollary 5.3 (Bridge structure). *The edge $\{(5, 7), (6, 6)\}$ is the unique edge of the distinguished-pair graph that bridges the four-cycle orbit V_6 and the observer orbit V_5 .*

6 The Four-Orbit Framework-Constant Theorem

6.1 The four-cycle orbit's anchors

The four-cycle orbit $V_6 = \{(5, 7), (7, 5), (1, 3), (3, 1)\}$ splits into two rs -pairs:

$$\text{Pair B (mother pair)} = \{(1, 3), (3, 1)\}, \quad (25)$$

$$\text{Pair A (daughter pair)} = \{(5, 7), (7, 5)\}. \quad (26)$$

The labels “mother” and “daughter” refer to the T_{half} relation: $T_{\text{half}}(1, 3) = (5, 7)$ and $T_{\text{half}}(3, 1) = (7, 5)$, so Pair B consists of the smaller- n_1 members and Pair A of the larger- n_1 members.

6.2 The intra-pair winding-matrix magnitudes

Definition 6.1 (Intra-pair winding-matrix magnitude). For an rs -pair $\{(n_1, n_2), (n_2, n_1)\}$, the intra-pair winding-matrix magnitude is

$$|M| = |n_1 \cdot n_1 - n_2 \cdot n_2| = |n_1^2 - n_2^2|. \quad (27)$$

Lemma 6.2 (Intra-pair magnitudes on V_6). *The intra-pair winding-matrix magnitudes of the two rs -pairs in V_6 are*

$$|M|_{\text{mother}} = L = 8, \quad |M|_{\text{daughter}} = (d - 1)L = 24. \quad (28)$$

Proof. For the mother pair $\{(1, 3), (3, 1)\}$:

$$|M|_{\text{mother}} = |1^2 - 3^2| = |1 - 9| = 8 = L. \quad (29)$$

For the daughter pair $\{(5, 7), (7, 5)\}$:

$$|M|_{\text{daughter}} = |5^2 - 7^2| = |25 - 49| = 24 = (d-1)L. \quad (30)$$

□

6.3 The pair sums and pair differences

Definition 6.3 (Pair sum and pair difference). For an rs -pair $\{(n_1, n_2), (n_2, n_1)\}$ with $N/4$ values x and y , the pair sum is $S = x + y$ and the pair difference is $D = |x - y|$.

Theorem 6.4 (Pair sums and pair differences). *The pair sums and pair differences of the two rs -pairs in V_6 are*

$$S_{\text{mother}} + S_{\text{daughter}} = \frac{2d-1}{d} (|M|_{\text{mother}} + |M|_{\text{daughter}}), \quad (31)$$

$$S_{\text{mother}} - S_{\text{daughter}} = \frac{2d-1}{L} (|M|_{\text{daughter}} - |M|_{\text{mother}}), \quad (32)$$

$$D_{\text{mother}} = \frac{(n_1 + n_2)_{\text{mother}}}{d}, \quad (33)$$

$$D_{\text{daughter}} = \frac{(n_1 + n_2)_{\text{daughter}}}{d}. \quad (34)$$

Proof. Substitute the intra-pair magnitudes from Lemma 6.2 and the coordinate sums

$$(n_1 + n_2)_{\text{mother}} = 1 + 3 = 4 = d, \quad (n_1 + n_2)_{\text{daughter}} = 5 + 7 = 12 = (d-1)d. \quad (35)$$

Then

$$S_{\text{mother}} + S_{\text{daughter}} = \frac{2d-1}{d} (L + (d-1)L) = \frac{2d-1}{d} \cdot dL = (2d-1)L = 7 \cdot 8 = 56,$$

$$S_{\text{mother}} - S_{\text{daughter}} = \frac{2d-1}{L} ((d-1)L - L) = \frac{2d-1}{L} \cdot (d-2)L = (2d-1)(d-2) = 7 \cdot 2 = 14,$$

$$D_{\text{mother}} = \frac{d}{d} = 1,$$

$$D_{\text{daughter}} = \frac{(d-1)d}{d} = d-1 = 3.$$

□

Corollary 6.5 (Pair sums). *The pair sums are*

$$S_{\text{mother}} = (d+1)(2d-1) = 35, \quad S_{\text{daughter}} = (d-1)(2d-1) = 21. \quad (36)$$

Proof. Solving the linear system

$$S_{\text{mother}} + S_{\text{daughter}} = 56,$$

$$S_{\text{mother}} - S_{\text{daughter}} = 14,$$

gives $S_{\text{mother}} = 35$ and $S_{\text{daughter}} = 21$. In framework-constant form, $S_{\text{mother}} = (d+1)(2d-1)$ and $S_{\text{daughter}} = (d-1)(2d-1)$. □

6.4 The $N/4$ values

Theorem 6.6 (Four-Orbit Framework-Constant Theorem). *The four $N/4$ anchors of the four-cycle orbit V_6 are the framework constants*

$$N/4(1, 3) = (d-1)(2d-1) = 21, \quad (37)$$

$$N/4(3, 1) = 2(L-1) = 14, \quad (38)$$

$$N/4(5, 7) = (d-1)d = 12, \quad (39)$$

$$N/4(7, 5) = (d-1)^2 = 9. \quad (40)$$

They satisfy the additive identities

$$N/4(7, 5) + N/4(5, 7) = N/4(1, 3) \quad (9 + 12 = 21), \quad (41)$$

$$N/4(7, 5) + N/4(3, 1) = H + 2d - 1 = 23 \quad (9 + 14 = 23). \quad (42)$$

Their sum is

$$N/4(1, 3) + N/4(3, 1) + N/4(5, 7) + N/4(7, 5) = 21 + 14 + 12 + 9 = 56 = dL \quad (43)$$

and their mean is

$$\frac{56}{4} = 14 = 2(L-1). \quad (44)$$

Proof. The base values are obtained from the pair sums and pair differences:

$$\text{base}(1, 3) = \frac{S_{\text{mother}} + D_{\text{mother}}}{2} = \frac{35 + 1}{2} = 18,$$

$$\text{base}(3, 1) = \frac{S_{\text{mother}} - D_{\text{mother}}}{2} = \frac{35 - 1}{2} = 17,$$

$$\text{base}(5, 7) = \frac{S_{\text{daughter}} + D_{\text{daughter}}}{2} = \frac{21 + 3}{2} = 12,$$

$$\text{base}(7, 5) = \frac{S_{\text{daughter}} - D_{\text{daughter}}}{2} = \frac{21 - 3}{2} = 9.$$

The δ values are determined by the T_{half} pair index k and the parity-block node:

$$\delta(1, 3) = 3, \quad \delta(3, 1) = -3, \quad \delta(5, 7) = 0, \quad \delta(7, 5) = 0. \quad (45)$$

Hence

$$N/4(1, 3) = 18 + 3 = 21 = (d-1)(2d-1),$$

$$N/4(3, 1) = 17 - 3 = 14 = 2(L-1),$$

$$N/4(5, 7) = 12 + 0 = 12 = (d-1)d,$$

$$N/4(7, 5) = 9 + 0 = 9 = (d-1)^2.$$

The additive identities and the sum and mean follow by arithmetic. □

Corollary 6.7 (Algebraic derivation of the sum). *The sum of the four-cycle orbit's $N/4$ anchors has the framework-constant form*

$$N/4(1, 3) + N/4(3, 1) + N/4(5, 7) + N/4(7, 5) = 2(d-1)(2d-1) + 14. \quad (46)$$

At $d = 4$, this evaluates to $2 \cdot 3 \cdot 7 + 14 = 42 + 14 = 56$.

Proof. Add the four framework-constant forms:

$$\begin{aligned}
& (d-1)(2d-1) + 2(L-1) + (d-1)d + (d-1)^2 \\
&= (d-1)[(2d-1) + d + (d-1)] + 14 \\
&= (d-1)[4d-2] + 14 \\
&= 2(d-1)(2d-1) + 14.
\end{aligned}$$

At $d = 4$: $2 \cdot 3 \cdot 7 + 14 = 56$. $\square$

Corollary 6.8 (Four-orbit as shadow of the Hodge sector). *The four-cycle orbit is a four-element shadow of the 56-dimensional Hodge sector at the self-paired eigenvalue: its four $N/4$ anchors sum to the sector dimension, and their mean is the multiplicity.*

7 Discussion

7.1 What the Klein structure gives

The Klein four-group $V = \langle rs, T_{\text{half}} \rangle$ organizes the fourteen-mode multiplet into six orbits at coordinate level. The orbit decomposition and its stabilizer classification are derived from the substrate. The six orbits are:

1. Four two-cycle orbits with pointwise stabilizer $\{e, rsT_{\text{half}}\}$: $V_1 = \{(0, 4), (4, 0)\}$, $V_2 = \{(6, 2), (2, 6)\}$, $V_3 = \{(1, 5), (5, 1)\}$, $V_4 = \{(3, 7), (7, 3)\}$.
2. One two-cycle orbit with pointwise stabilizer $\{e, rs\}$: the observer orbit $V_5 = \{(2, 2), (6, 6)\}$.
3. One four-cycle orbit with trivial pointwise stabilizer: $V_6 = \{(5, 7), (7, 5), (1, 3), (3, 1)\}$.

The edge $\{(5, 7), (6, 6)\}$ is the unique bridge between the four-cycle orbit and the observer orbit. This is the framework's minimal connective structure between its two most distinctive orbits.

7.2 Refinement of the D_4 orbit partition

The D_4 orbit partition of the fourteen-mode multiplet is

$$\begin{aligned}
\mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\
\mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\
\mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}.
\end{aligned}$$

The Klein orbit partition refines the D_4 orbit partition:

- $\mathcal{O}_0$ is the single Klein orbit $V_1 = \{(0, 4), (4, 0)\}$.
- $\mathcal{O}_1$ decomposes into three Klein orbits: $V_3 = \{(1, 5), (5, 1)\}$, $V_4 = \{(3, 7), (7, 3)\}$, and $V_6 = \{(5, 7), (7, 5), (1, 3), (3, 1)\}$.
- $\mathcal{O}_2$ decomposes into two Klein orbits: $V_2 = \{(6, 2), (2, 6)\}$ and $V_5 = \{(2, 2), (6, 6)\}$.

7.3 Relation to the observer pair

The coordinate pair $\{(2, 2), (6, 6)\}$ of HCSM-26 is the Klein orbit V_5 . Its pointwise stabilizer is $\{e, rs\}$, the subgroup generated by the chiral reflection. This is consistent with the Chirality Selection Theorem of HCSM-26: the coordinate pair is the unique fixed locus of rs on the fourteen-mode multiplet.

The four-cycle orbit V_6 has trivial pointwise stabilizer. Its swap partner with the observer orbit is the coordinate $(6, 6)$. The two most distinctive orbits of the Klein structure are connected by the edge $\{(5, 7), (6, 6)\}$.

8 Inputs and Derived

8.1 Derived from the substrate

The following results are proved without reference to the $N/4$ anchors or to particle identifications:

- (1) The Klein structure $rs^2 = T_{\text{half}}^2 = e$ and $[rs, T_{\text{half}}] = 0$ (Lemma 3.1).
- (2) The orbit decomposition into five size-2 orbits and one size-4 orbit (Theorem 3.2).
- (3) The pointwise stabilizer classification of each orbit (Theorem 4.2 and Corollary 4.3).
- (4) The ten edges of the distinguished-pair graph and the unique inter-orbit edge $\{(5, 7), (6, 6)\}$ (Theorem 5.2).
- (5) The intra-pair winding-matrix magnitudes $|M|_{\text{mother}} = L$ and $|M|_{\text{daughter}} = (d-1)L$ (Lemma 6.2).
- (6) The pair-sum and pair-difference identities (Theorem 6.4).
- (7) The pair sums and pair differences (Corollary 6.5).

8.2 Derived given the substrate-level inputs

The following result is derived from the substrate-level inputs listed in Section 2.8:

- (8) The Four-Orbit Framework-Constant Theorem (Theorem 6.6).
- (9) The algebraic derivation of the sum (Corollary 6.7).
- (10) The four-orbit-as-shadow statement (Corollary 6.8).

8.3 Inputs

Input 1. SM content. The Standard Model particles and their observed properties, as used to identify the fourteen modes.

Input 2. Physical identifications. The identifications $A_1 \leftrightarrow \text{color}$, $B_1 \leftrightarrow \text{weak isospin}$.

Input 3. Identification thesis. The assignment of each of the fourteen modes at $\lambda = 4$ to a specific Standard Model particle, together with the $N/4$ mass label attached to each.

Input 4. Framework constants. $d = 4$, $N = 64$, $L = 8$, $H = 16$.

8.4 What is not claimed

We do not claim to derive the identification thesis from the substrate. The assignment of the fourteen modes to the fourteen particles, and the $N/4$ labels attached to each, are the framework’s empirical anchors.

We do not claim that the Four-Orbit Framework-Constant Theorem is HCSM-native without qualification. It is derived from the substrate-level inputs of Section 2.8.

We do not claim that the naming of the coordinates $(5, 7)$ and $(6, 6)$ as dark and t is derived. The coordinates and the edge structure are derived; the naming is anchored.

The framework’s claim is that it has fewer empirical anchors than the Standard Model, not that it has none.

9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script constructs the fourteen-mode multiplet, the diagonal reflection rs , the half-period translation T_{half} , the Klein four-group $V = \langle rs, T_{\text{half}} \rangle$, the orbit decomposition, the pointwise stabilizers, the distinguished-pair graph, the intra-pair winding-matrix magnitudes, the pair sums and pair differences, and the four-cycle orbit’s anchors from HCSM-native definitions, and verifies:

- (1) The Klein four-group presentation $rs^2 = T_{\text{half}}^2 = e$ and $[rs, T_{\text{half}}] = 0$.
- (2) The orbit decomposition into four two-cycle orbits with stabilizer $\{e, rsT_{\text{half}}\}$, one two-cycle orbit with stabilizer $\{e, rs\}$, and one four-cycle orbit with trivial stabilizer.
- (3) The pointwise stabilizer classification of each orbit, computed purely from coordinates.
- (4) The unique inter-orbit edge $\{(5, 7), (6, 6)\}$ in the distinguished-pair graph.
- (5) The intra-pair winding-matrix magnitudes $|M|_{\text{mother}} = L$ and $|M|_{\text{daughter}} = (d - 1)L$.
- (6) The pair-sum and pair-difference identities.
- (7) The four-cycle orbit’s $N/4$ anchors and their framework-constant forms.

- (8) The additive identities $9 + 12 = 21$ and $9 + 14 = 23$.
- (9) The sum 56 and the mean 14.

Running `python HCSM27_verification.py` reproduces every numerical claim in this paper.

10 Conclusion

This paper has characterized the Klein orbit structure of the fourteen-mode multiplet in the Hodge Complex Standard Model.

Three theorems are fully derived from the substrate and do not use the $N/4$ anchors. The *Klein Orbit Theorem* establishes that the Klein four-group generated by the diagonal reflection and the half-period translation acts on the multiplet with orbit decomposition consisting of five size-2 orbits and one size-4 orbit. The *Kernel Classification Theorem* establishes that the pointwise stabilizer of each orbit is one of the three non-trivial subgroups of the Klein group. The *Swap-Pair Bridge Theorem* establishes that the edge $\{(5, 7), (6, 6)\}$ is the unique inter-orbit edge of the distinguished-pair graph.

One theorem is derived from the intra-pair winding-matrix magnitudes, which are coordinate arithmetic on the four-cycle orbit. The *Four-Orbit Framework-Constant Theorem* establishes that the four-cycle orbit’s four $N/4$ values are simultaneously framework constants, satisfy two additive identities within the orbit, and sum to fifty-six, the dimension of the Hodge sector at the self-paired eigenvalue; their mean equals fourteen, the multiplicity.

The Klein structure organizes the fourteen-mode multiplet in a way that refines the D_4 orbit partition and brings the framework-constant arithmetic of the four-cycle orbit into focus. The observer pair of HCSM-26 is the orbit V_5 of this structure, and the framework-constant orbit is V_6 .

The framework’s claim is that the Standard Model is what the substrate looks like when you observe it. The identification of the fourteen modes with the fourteen particles is the empirical thesis; the structural relations among the $N/4$ anchors are theorems given the thesis. The framework’s strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

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